



pens

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Materi : Praktikum 1, Pengenalan Simulink

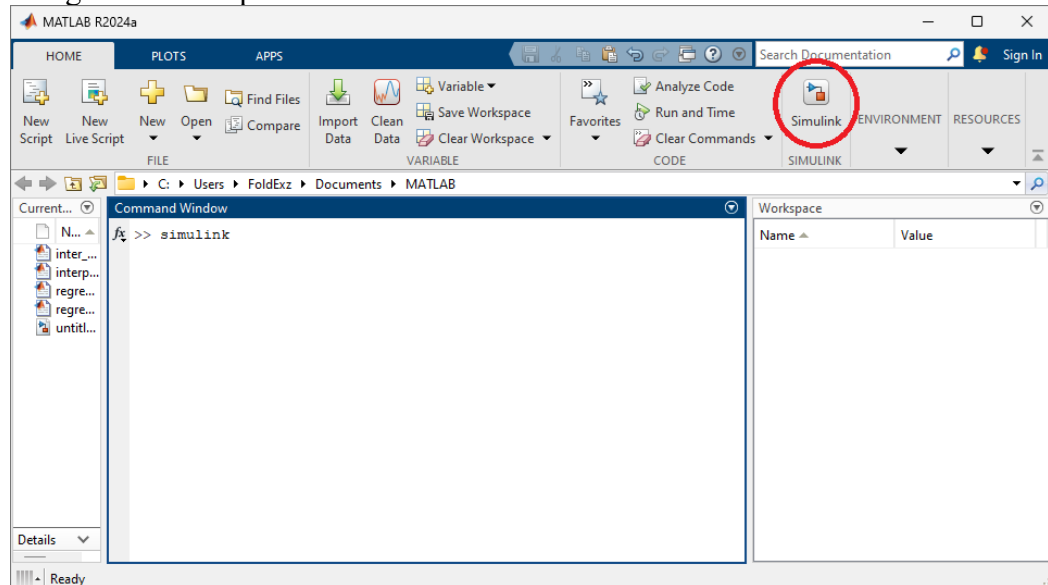
Tanggal : Rabu, 3 September 2025



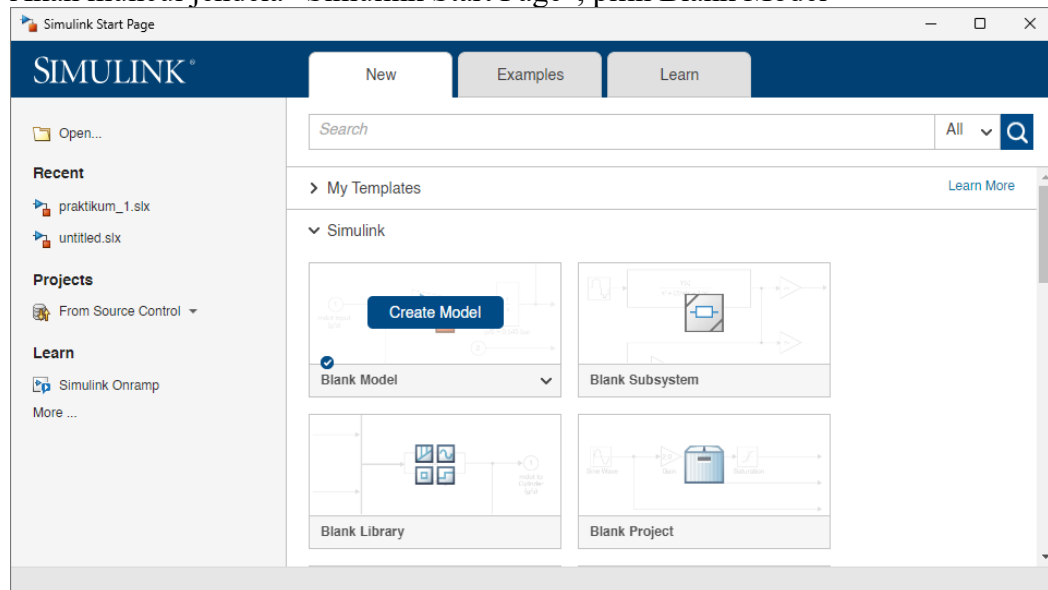
III. Experimental Setup

1. Permulaan

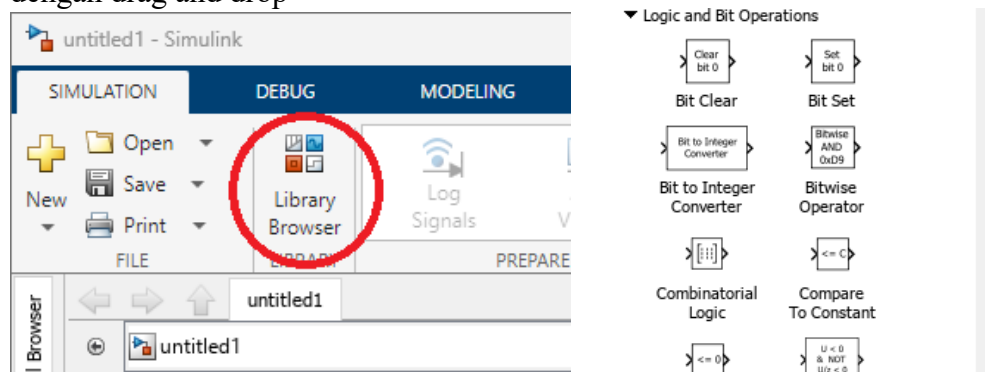
- a. Jalankan simulink dengan mengetikkan “simulink” pada Command Window atau meng-klik icon seperti dibawah ini



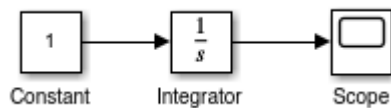
- b. Akan muncul jendela “Simulink Start Page”, pilih Blank Model



- c. Klik pada “Library Browser” dan pilih blok sesuai kebutuhan ke lembar kerja dengan drag and drop



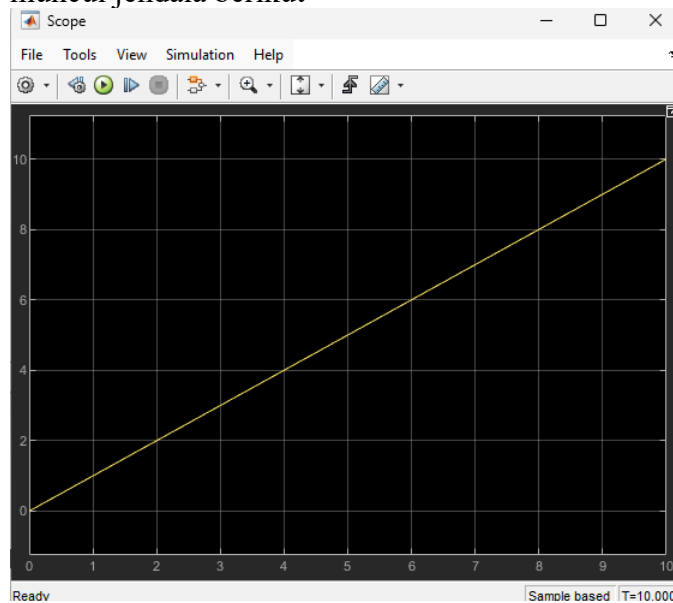
d. Buatlah model simulink seperti berikut:



e. Jalankan simulasi dengan klik icon start



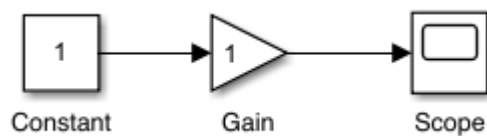
f. Untuk melihat data hasil simulasi, double klik pada simbol “Scope” sehingga muncul jendela berikut



2. Percobaan

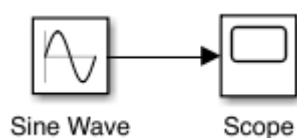
Buatlah model simulink seperti berikut:

a.



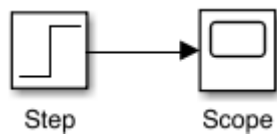
Model ini dapat menghasilkan sinyal konstan yang bernilai 1 kemudian dikalikan dengan faktor penguat bernilai 1, dan hasil keluarannya ditampilkan pada Scope.

b.



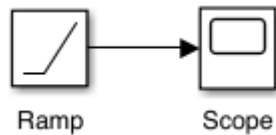
Model ini menghasilkan sinyal berbentuk gelombang sinus secara periodik, dan hasil keluarannya ditampilkan pada Scope.

c.



Model ini menghasilkan sinyal loncatan (step) dari nilai awal ke nilai akhir pada waktu tertentu, dan hasil keluarannya ditampilkan pada Scope.

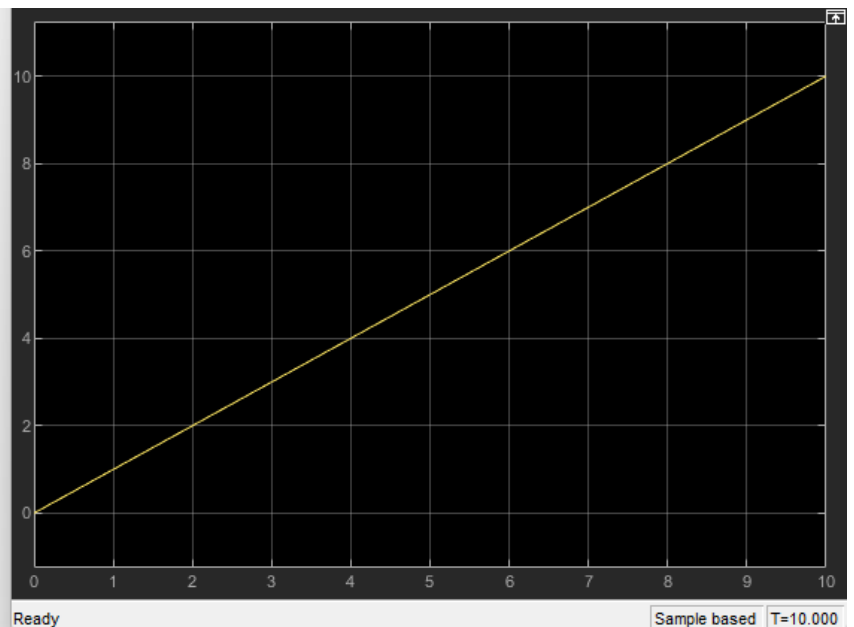
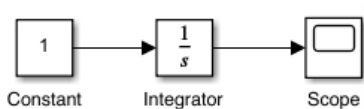
d.



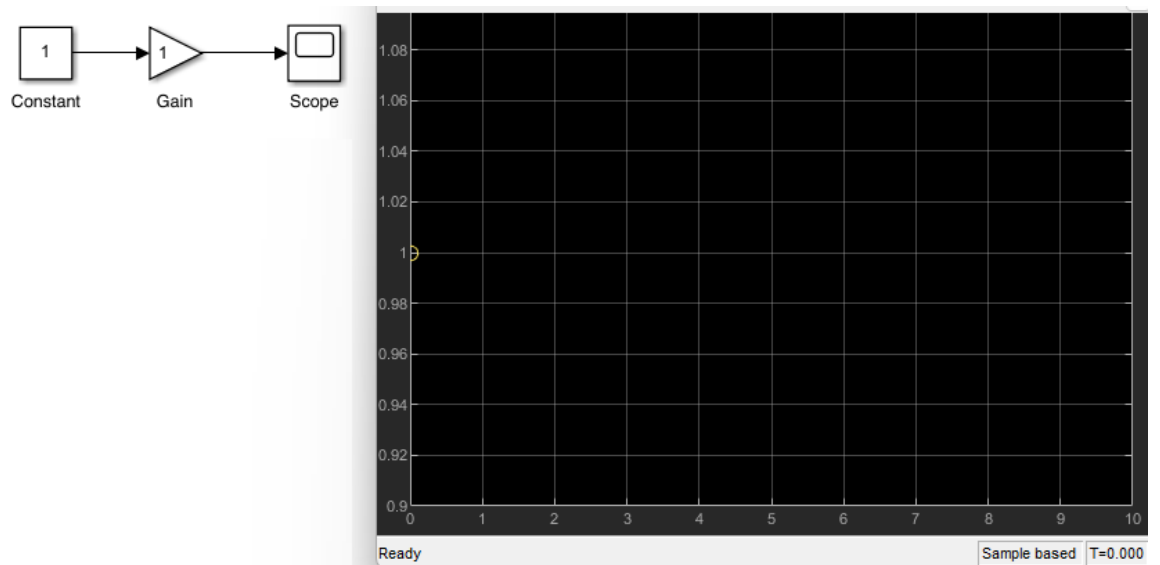
Model ini menghasilkan sinyal ramp yang naik secara linear terhadap waktu, dan hasil keluarannya ditampilkan pada Scope.

IV. Hasil Pengujian

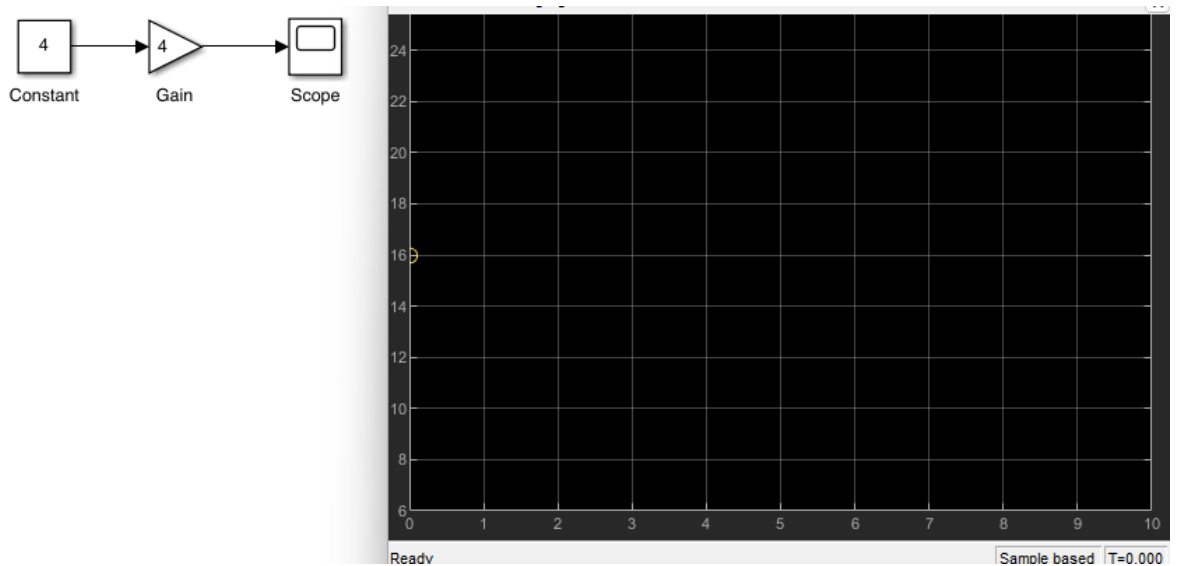
Hasil pada prosedur



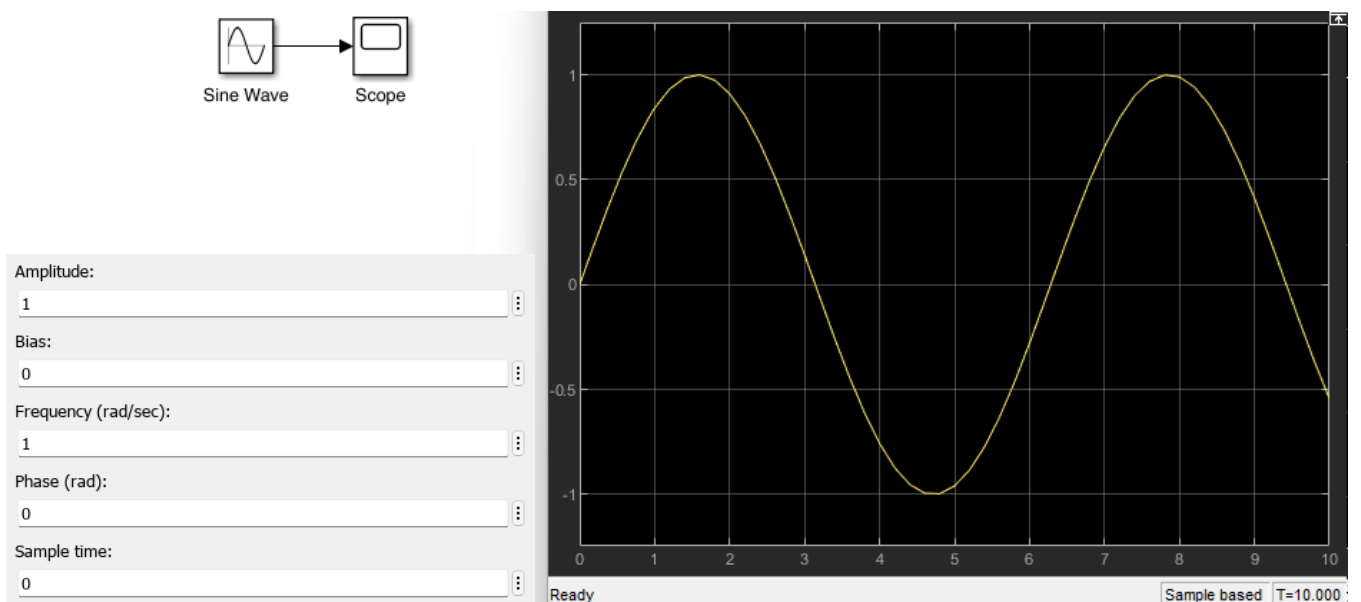
A. Percobaan model a dengan $T = 0s$; Gain = 1; Constant = 1



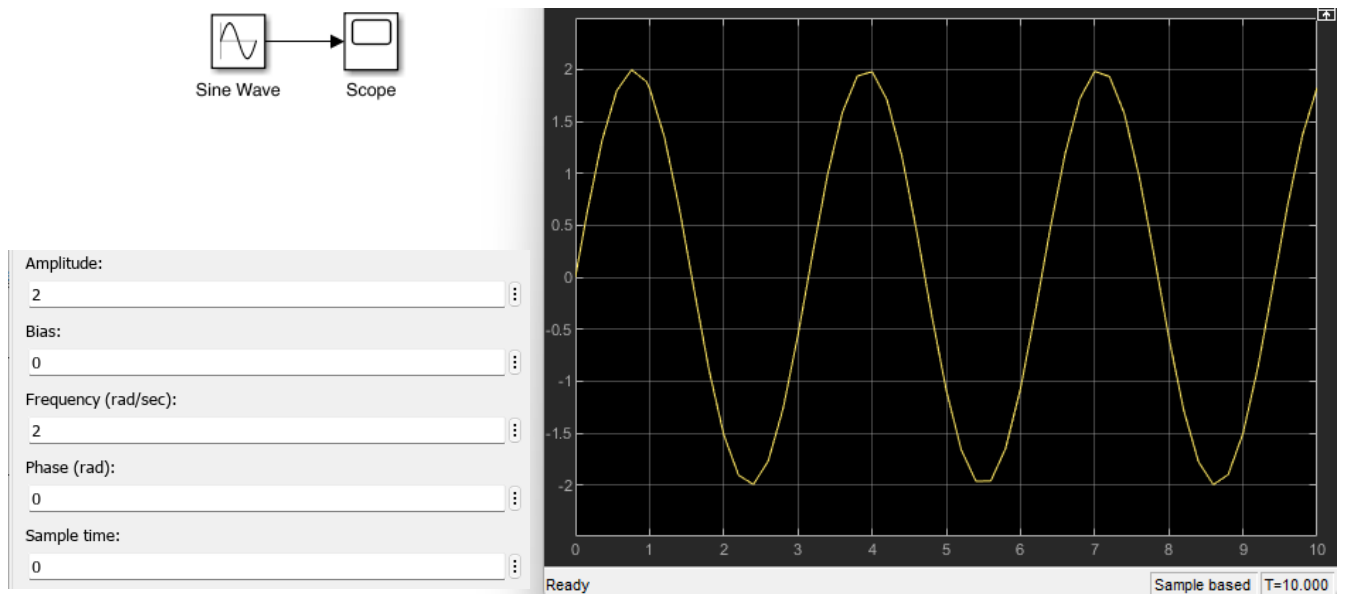
Pengubahan parameter, Gain = 4 ; Constant = 4



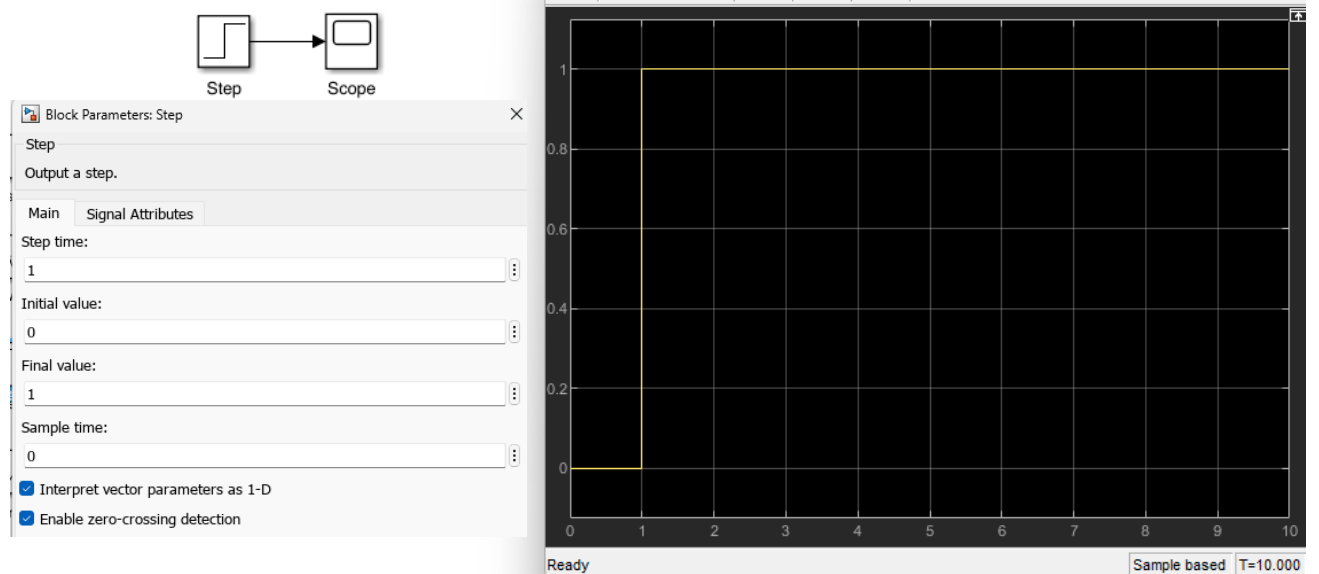
B. Percobaan model b dengan $T = 10.0s$; A = 1; Freq = 1



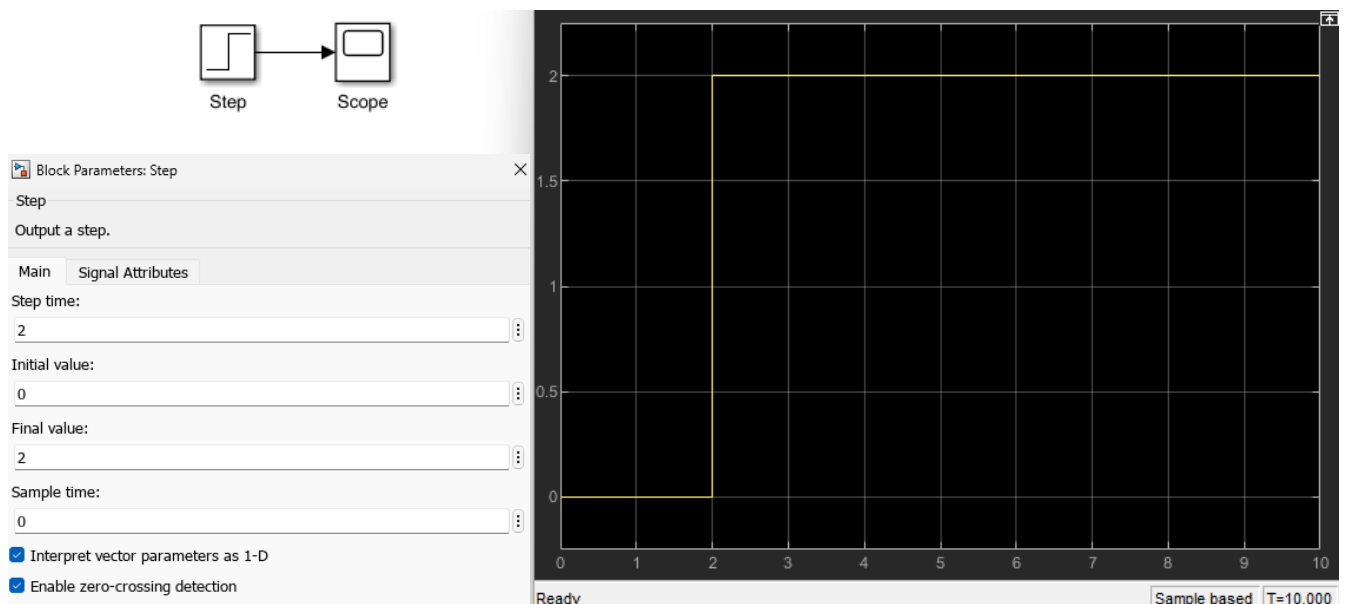
Pengubahan parameter, $A = 2$; Freq = 2



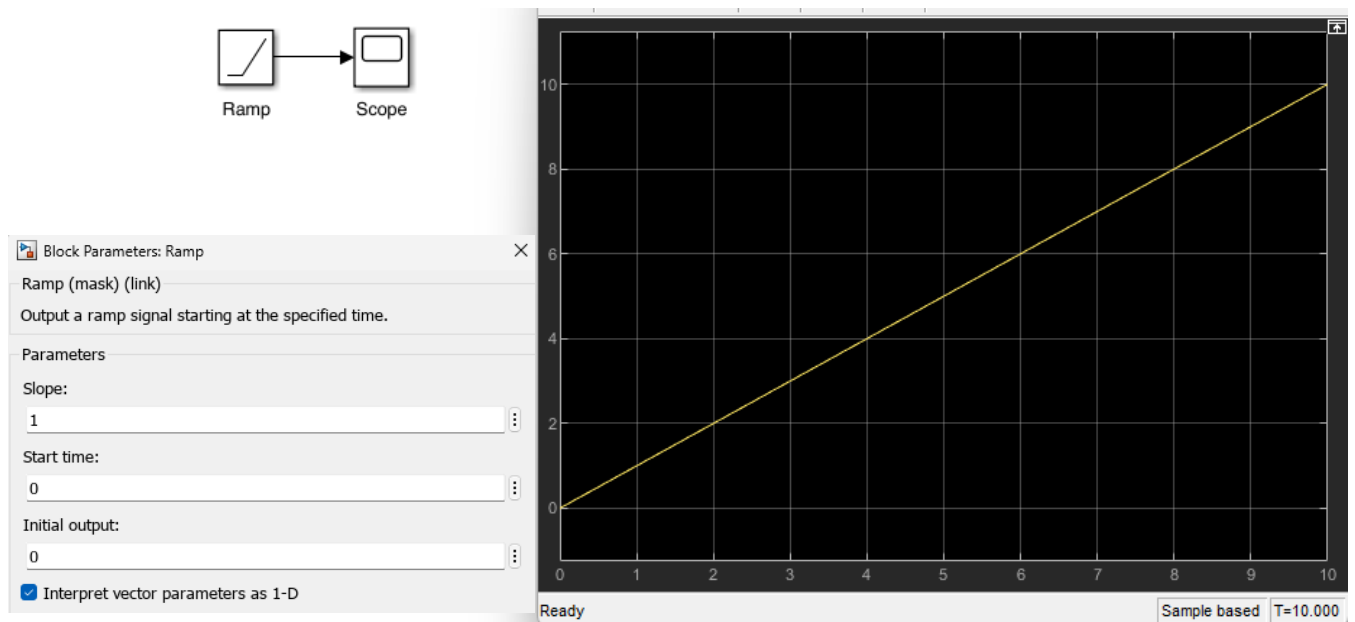
C. Percobaan model c dengan $T = 10.0s$; Step Time = 1; Final Value = 1



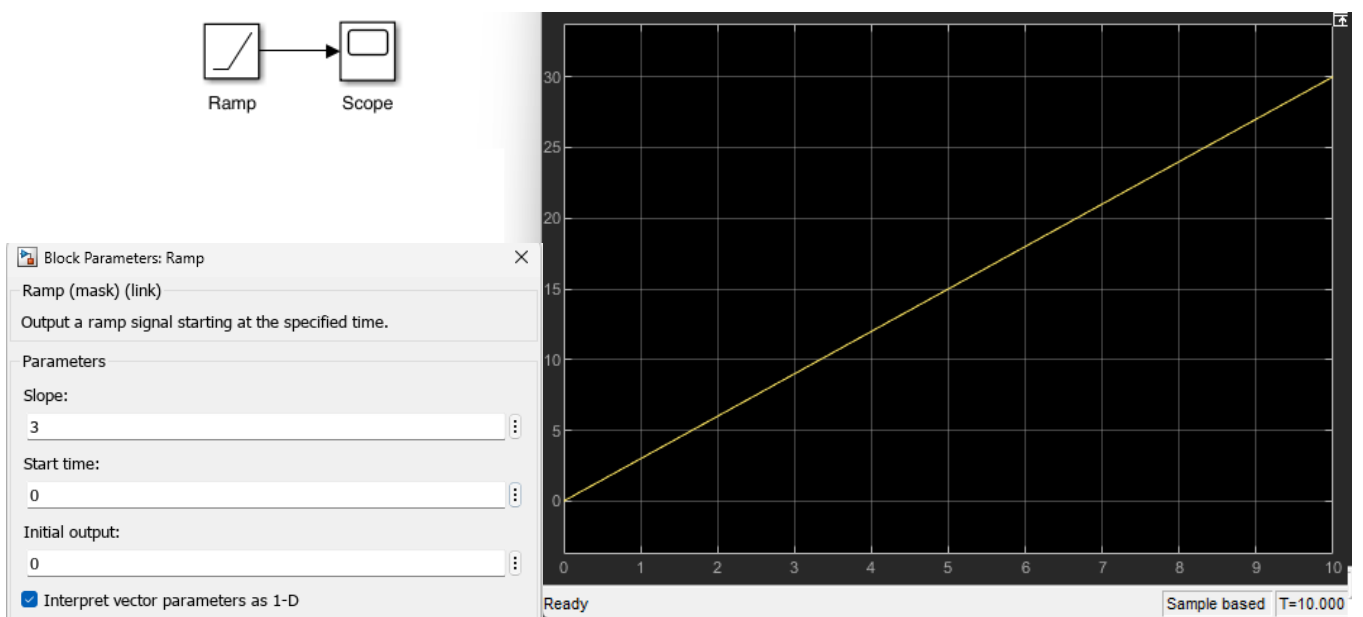
Pengubahan parameter, Step Time = 2; Final Value = 2



D. Percobaan model d dengan $T = 10.0\text{s}$; Slope = 1



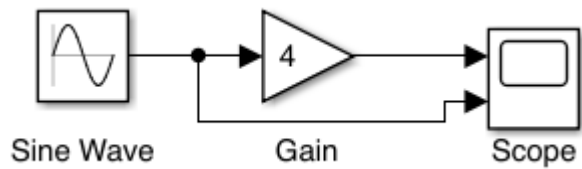
Pengubahan parameter, Slope = 3



TUGAS

Buatlah Model simulink yang memnunjukkan proses terjadinya :

1. Penguatan sinyal sinus



Nilai sin:

Amplitude:
1

Bias:
0

Frequency (rad/sec):
0.5*pi 1.5708

Phase (rad):
0

Sample time:
0

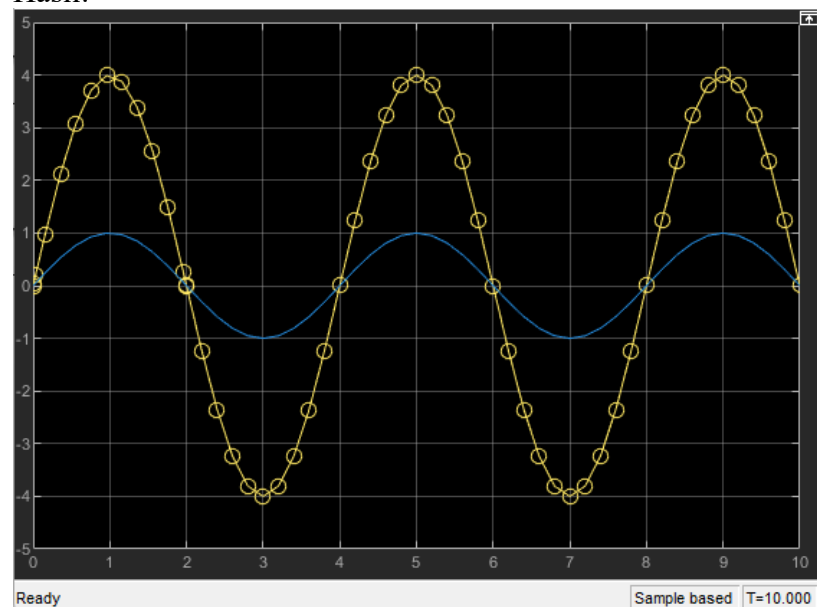
☒ Interpret vector parameters as 1-D

Nilai gain:

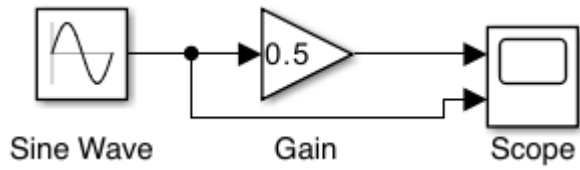
Gain:
4

Multiplication: Element-wise(K.*u)

Hasil:



2. Pelemahan sinyal sinus



Nilai sin :

Amplitude: 1

Bias: 0

Frequency (rad/sec): $0.5 \cdot \pi \cdot 4$ 6.2832

Phase (rad): 0

Sample time: 0

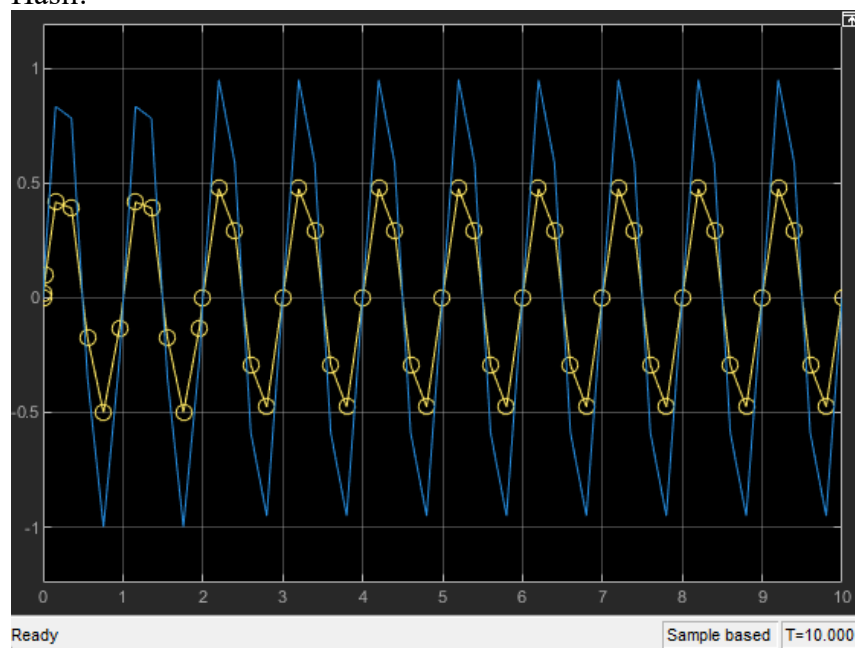
☒ Interpret vector parameters as 1-D

Nilai gain:

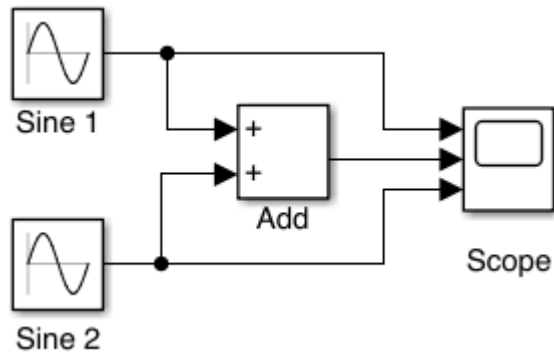
Gain: 0.5

Multiplication: Element-wise($K \cdot u$)

Hasil:



3. Penjumlahan dua sinyal sinus



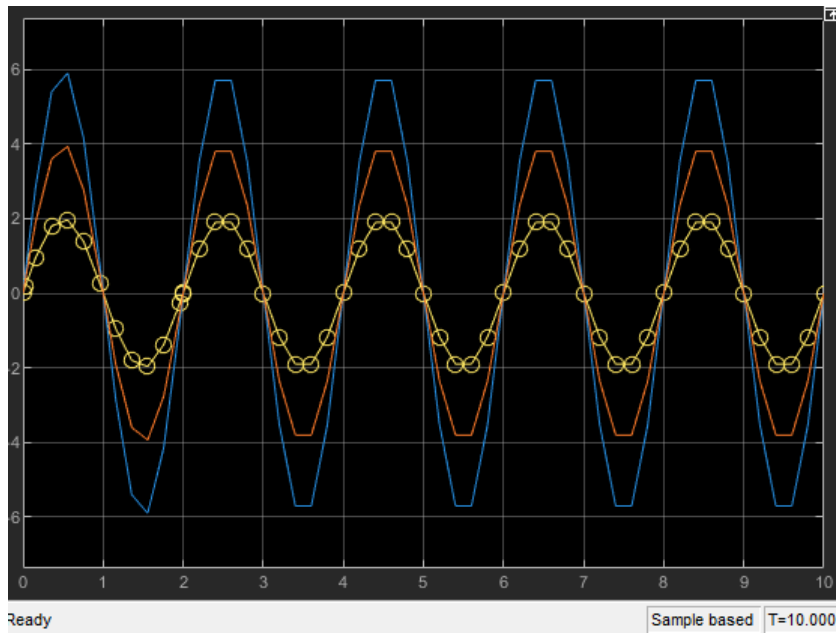
Nilai sin 1:

Amplitude:	2
Bias:	0
Frequency (rad/sec):	pi
Phase (rad):	0
Sample time:	0
☒ Interpret vector parameters as 1-D	

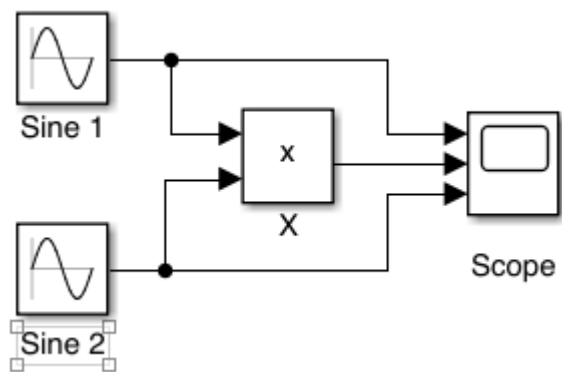
Nilai sin 2:

Amplitude:	4
Bias:	0
Frequency (rad/sec):	pi
Phase (rad):	0
Sample time:	0
☒ Interpret vector parameters as 1-D	

Hasil:



4. Perkalian dua sinyal sinus



Nilai sin 1:

Amplitude:
2

Bias:
0

Frequency (rad/sec):
pi

Phase (rad):
0

Sample time:
0

☒ Interpret vector parameters as 1-D

Nilai sin 2:

Amplitude:
4

Bias:
0

Frequency (rad/sec):
 π

Phase (rad):
0

Sample time:
0

☒ Interpret vector parameters as 1-D

Hasil:

